United International **University (UIU)**

Dept. of Computer Science & Engineering (CSE)

Mid-term Exam: : Trimester: Summer 2025

Course Code: CSE 1111, Course Title: STRUCTURED PROGRAMMING LANGUAGE

Time: 1 hour 30 min Total Marks: 30

(Unfair means in the exam will be taken seriously as per UIU disciplinary rules)

Answer all the Questions. Numerical values in the right margin indicate full marks.

1 a) Show the **manual tracing** of the following code **[3]**

```
#include<stdio.h>
int main() {
int sum=0, i=10;
    sum=sum+i++;
    i=i+1;
    ++i;
    i--;
    sum+=i;
    printf("sum=%5d i=%d", sum, i);
    return 0;
}
```

b) Find **output** of the following code segment **[3]**

```
int a=3, b=4, c=5, result;
  result = c % b + a;
  printf("result = %d\n", result);
  if (result >= 0 && result < 10) {
       printf("a =  %d\n", a);
  }
  else if (result >= 5) {
       printf("b =  %d\n", b);
  }
  else printf("a = %d\n", a);
```

2 a) **Rewrite** the code segment using "**if … else**" without changing the logical meaning **[4]**

```
int num, sum=0;
scanf("%d",&num);
switch(num){
   case 1:
   case 2: sum=sum+num;
   case 3:
           printf("%d\n", sum); break;
   default:
           printf("%d\n", num);
}
```

b) **Manually trace** the following code segment **[4]**

```
int a=14, b=4, R, Q;
   Q=0;
   R=a;
   while((a-b)>=0){
        R=a-b;
        ++Q;
        a=R;
   }
```

		`printf("Q=%d\n R=%d", Q, R);`	

3	a)	Draw a **flow chart** for the given code segment
		`int i, num=2, sum=0, count=0;`
		`do{`
		`      for(i=1; i<=num/2; i++){`
		`        if (num%i==0){`
		`           sum=sum+i;`
		`        }`
		`      }`
		`      if(num==sum){`
		`         printf("%d", num);`
		`         ++count;`
		`         sum=0;`
		`      }`
		`      ++num;`
		`}while(count<=3);`
			4

	b)	Write **a complete program** that finds the moving average from all the input positive integer numbers from keyboard until n positive numbers are found. For example, for n=3	4

Input Number	Average Calculation	Output
16	16/1=16.0	16.0
17	(16+17)/2=16.5	16.5
-18	-	16.5
20	(16+17+20)/3=17.67	17.67
-24	-	17.67

4	a)	**Manually trace** the given code segment below.
		`int F[5]={0};`
		`int i;`
		`F[0]=1;`
		`F[1]=1;`
		`for(i=1; i<=3; i++){`
		` F[i+1]=F[i]+F[i-1];`
		` printf("F[%d]=%d ", i-1,F[i-1]);`
		` printf("F[%d]=%d ", i,F[i]);`
		` printf("F[%d]=%d\n",i+1,F[i+1]);`
		`}`
			4

	b)	Write **a complete program** that finds sum of all the elements in a row of a two-dimensional array of NxN and stores all the sums in a one-dimensional array. For example,	4

N	**2D Array**	**Sums for all Rows**
3	1 4 7 2 8 3 5 6 0	**{1+4+7, 2+8+3, 5+6+0} = {12, 13, 11}**